

Tomogrations — pipeline reference

Every command: what it reads, the exact filenames it looks for, what it writes and where. Generated from the app's own stage definitions.

Notation `PositionNNN` = a tilt series · `<apx>` = pixel size to TWO decimals (`--angpix 10` → `10.00Apx`) · `<processing>` = `warp_tiltseries/` for a trunk run, or `jobs/J###/` for a job.

The job model — jobs/J###/

Every run started from the CARD VIEW is a **job** with its own processing directory `jobs/J###/`. WarpTools' `--input_processing / --output_processing` (available on every command) redirect where a run READS its metadata and WRITES its results, so a job reads its parent's directory and writes its own. That is what lets two variants of the same step coexist instead of overwriting one another.

Bulk data (tilt stacks, raw movies) is resolved through the `.settings` file and is NOT duplicated — a branch only costs the per-series XML (~0.4 MB each). Products that a step genuinely creates (tomograms, subtomograms) live only in the job that made them.

Trunk runs (the classic ▶ Run) skip all this and write to the shared conventional folders above. Both are fine; jobs are what you want when you're comparing variants.

1. DATA PREP

Rename EER + MDOC

```
bash ml_batch_rename_eer_mdoc_mrc_warp_auto.sh
```

Renames Tomo5 beam-shift names (`Position_1_2`) to Warp form (`Position001`). Writes `listfile_<root>.txt` mapping old->new and fixes mdoc dates to `yy-mm-dd` (required by Warp).

READS

dirs: .

<code><source_dir>/*.eer</code>	raw movies from the microscope
<code><source_dir>/*.mdoc</code>	one per tilt series (SerialEM/Tomo5)

WRITES

dirs: mdocs, frames

<code><source_dir>/PositionNNN*.eer</code>	renamed in place
<code><source_dir>/PositionNNN.mdoc</code>	renamed in place
<code>listfile_Position.txt</code>	map: original Tomo5 name -> PositionNNN

Watch out

- RENAME FIRST, then Sort files. The script globs `*.mdoc` + matching `*.eer` together in ONE folder; sorting first breaks that pairing.
- `*_override.mdoc` files are quarantined to `mdocs/bad/` (they would take bogus `PositionNNN` slots).

IMOD → Warp key

```
python ml_imodtowarpkey_generator_warp_auto.py
```

Builds the IMOD->Warp tilt-number key. mdoc ZValue blocks are in dose-symmetric acquisition order; 3dmod shows tilts in angle order. The key translates between them.

READS

dirs: mdocs

<code>mdocs/PositionNNN.mdoc</code>	one sample mdoc, for the tilt order
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WRITES

dirs: .

<code>new_imod_conv_key.txt</code>	IMOD tilt-index -> mdoc-section key
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Watch out

- Feeds `remake_mdocs`, which uses it to translate IMOD tilt numbers.

Inspect tilt stacks

(no command – an in-app tool)

Visually review each tilt series (thumbnail grid + 3dmod) and mark bad tilts (IMOD #, ranges OK e.g. 1,4-12,48) or whole series for exclusion before remaking the mdocs.

READS

dirs: Thumbnails, mdocs

Thumbnails/*.mrc	per-series montage (one per tilt series)
mdocs/*.mdoc	to list the series

WRITES

dirs: mdocs

exclusion_list.txt	manual section: `PositionNNN<tab>1,4-12`
mdocs/bad/	quarantined whole series (mdoc moved here)
frames/bad/	quarantined whole series (.eer moved here)

Watch out

- A BARE `PositionNNN` line (no tilt numbers) = drop the WHOLE series.
- Tilt ranges like `4-12` are expanded before remake_mdocs (its sed can't parse ranges).

Remake MDOCs

bash ml_batch_remake_mdocs_warp_auto.sh

Removes excluded tilts from mdocs and renumbers remaining ZValue blocks contiguously, then fixes mdoc date format.

READS

dirs: mdocs

mdocs/*.mdoc	the series to rewrite
exclusion_list.txt	which tilts/series to drop
new_imod_conv_key.txt	IMOD -> mdoc section mapping

WRITES

dirs: mdocs

mdocs/PositionNNN.mdoc	rewritten without the excluded tilts
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Watch out

- Needs ABSOLUTE paths (it cd's into mdocs/ and reads the lists by name).

2. GAIN

Gain: convert

module load eman && e2proc3d.py

Converts a camera .gain reference to .mrc via EMAN2.

READS

dirs: gains

gains/<name>.gain	the raw gain reference (name varies per dataset)
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WRITES

dirs: gains

gains/original_gain.mrc	gain as MRC
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Watch out

- The default `gains/original\_gain` usually does NOT exist — the real gain file is auto-detected; check the name.

Gain: reciprocal

module load eman && e2proc2d.py

Computes the reciprocal gain that Linux WarpTools expects.

READS

dirs: gains

gains/original_gain.mrc	from gain_convert
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WRITES

dirs: gains

gains/gain_reciprocal.mrc	1/gain — this is what create_settings uses
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Flags `--process math.reciprocal` on

Watch out

- Linux WarpTools wants the reciprocal; using the plain gain double-applies the correction.

3. FRAMESERIES

Frameseries settings

WarpTools create_settings

Writes the Warp frame-series .settings file.

READS

dirs: frames, gains

frames/*.eer	just to locate/describe the data
gains/gain_reciprocal.mrc	recorded as the gain path

WRITES

dirs: warp_frameseries

warp_frameseries.settings	the frameseries settings file
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Flags `--folder_data` frames `--folder_processing` warp_frameseries `--output` warp_frameseries.settings `--extension` *.eer `--angpix` 1.57 `--gain_path` gains/gain_reciprocal.mrc `--exposure` 3.5 `--eer_ngroups` 10

Watch out

- Writes NO per-movie data — it only records paths/pixel size/exposure.
- eer_ngroups IS the correct flag on this Warp 2.0 build.

Motion + CTF

WarpTools fs_motion_and_ctf

Per-frame-series motion correction + CTF estimation.

READS

dirs: frames, warp_frameseries

frames/*.eer	the raw movies
warp_frameseries.settings	paths, angpix, gain, exposure

WRITES

dirs: warp_frameseries

warp_frameseries/<movie>.xml	per-movie motion + CTF metadata
warp_frameseries/average/<movie>.mrc	aligned average (--out_averages)
warp_frameseries/averagehalves/<movie>.mrc	half-averages (--out_average_halves)

Flags `--settings` warp_frameseries.settings `--m_grid` 1x1x3 `--c_grid` 2x2x1 `--m_range_min` 500 `--m_range_max` 10 `--m_bfac` -500 `--c_range_max` 7 `--c_defocus_max` 8 `--out_averages` on `--out_average_halves` on `--device_list` 0 `--perdevice` 2

Watch out

- MUST pass --out_averages (and --out_average_halves). Without them no aligned averages are written and ts_import fails for EVERY series with 'does not have an aligned average result'.
- Idempotent: re-running skips movies that already have XML.

4. TILT SERIES

Tilt-series settings

WarpTools create_settings

Writes the Warp tilt-series .settings file.

READS

dirs: mdocs

(nothing yet — tomostar/ may be empty)

WRITES

dirs: warp_tiltseries

warp_tiltseries.settings	the tilt-series settings file
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Flags `--folder_data` tomostar `--folder_processing` warp_tiltseries `--output` warp_tiltseries.settings `--extension` *.tomostar `--angpix` 1.57 `--gain_path` gains/gain_reciprocal.mrc `--exposure` 3.5 `--tomo_dimensions` 4096x4096x3088

Watch out

- Run BEFORE ts_import. Seeing '0 files found / tomostar not found' here is EXPECTED — tomostar/ doesn't exist yet.

Import tilt series

WarpTools ts_import

Builds .tomostar files by pairing mdocs with frameseries.

READS

dirs: mdocs, warp_frameseries

mdocs/*.mdoc	tilt angles, order, exposure
warp_frameseries/<movie>.xml	the per-movie results + aligned averages

WRITES

dirs: tomostar, warp_tiltseries

tomostar/PositionNNN.tomostar	one per tilt series — the item list r
warp_tiltseries/PositionNNN.xml	per-series metadata (created here)

Flags `--mdocs` mdocs `--frameseries` warp_frameseries `--tilt_exposure` 3.5 `--min_intensity` 0 `--dont_invert` on `--output` tomostar `--override_axis`

Watch out

- CANNOT be scoped with `--input_data` (only `--mdocs` / `--pattern`), which is why subset work uses a separate project folder.

Build tilt stacks

WarpTools ts_stack

Builds aligned tilt stacks (.st) per tilt series for AreTomo.

READS

dirs: warp_tiltseries, tomostar

tomostar/*.tomostar	the series
warp_frameseries/average/*.mrc	the aligned averages

WRITES

dirs: warp_tiltseries/tiltstack

warp_tiltseries/tiltstack/PositionNNN/PositionNNN.st	the tilt stack
warp_tiltseries/tiltstack/PositionNNN/*.rawtilt	tilt angles

Flags `--settings` warp_tiltseries.settings `--angpix` `--device_list` 0 `--perdevice` 2

Watch out

- Run `ts_import` first; a missing mdoc entry stalls a stack.

5. ALIGNMENT

Align (AreTomo2)

bash

Marker-free tilt-series alignment (+optional recon). Writes .xf/.tlt per series under <output>/lmod/.

READS

dirs: warp_tiltseries/tiltstack

warp_tiltseries/tiltstack/*/* the tilt stacks
*.st

WRITES

dirs: aretomo_output

aretomo_output[-vN]/PositionNNN/	per-series AreTomo output
aretomo_output[-vN]/lmod/PositionNNN/PositionNNN.xf	the ALIGNMENT — this is the only thing Warp needs
aretomo_output[-vN]/PARAMETERS.txt	what was run

Flags `ARETOMO_GPUS` 0 1 2 3 `ARETOMO_JOBS_PER_GPU` 1 `ARETOMO_ALIGNZ` 670 `ARETOMO_VOLZ` 3088 `ARETOMO_OUTBIN` 8 `ARETOMO_DARKTOL` 0.000001 `ARETOMO_TILTCOR` 0 `ARETOMO_FLIPVOLZ` 1 `ARETOMO_WBP` 1 `ARETOMO_TILTAXIS` `ARETOMO_PATCH` `ARETOMO_ALIGN` 1 `ARETOMO_BIN` /ceph/groups/structbio/Programs/AreTomo2/AreTomo2 `ARETOMO_CUDA_LIB`

Watch out

- Auto-VERSIONS: each run makes a NEW aretomo_output-vN folder. A killed run leaves a PARTIAL folder — a later version can have FEWER .xf than an earlier one. Check: for d in aretomo_output*; do echo "\$d: \$(find \$d -name '*.xf'|wc -l)"; done
- Set `ARETOMO_VOLZ=0` for ALIGNMENT-ONLY (skips the slow tomogram; you only need the .xf).

Refine alignment (train)

bash

Deep-learning REFINEMENT of an EXISTING alignment by TRAINING a model on this set (warpem/miss-alignment). Does NOT align raw stacks — the docs state 'miss-alignment starts from an initially coarse aligned dataset', so coarse-align FIRST (AreTomo → ts_import_alignments).

READS

dirs: warp_tiltseries, warp_tiltseries/tiltstack

warp_tiltseries/PositionNNN. reads AND REWRITES these in place
xml

warp_tiltseries/tiltstack/ the stacks

Flags MA_TRAINING_DEVICES 0 MA_RECON_DEVICES 0,0,0 MA_DATALOADERS 5 MA_POOL_SIZE 2000 MA_START_ITER MA_PREPARE_STACKS 10.0
MA_CONDA_ENV miss-alignment

WRITES

dirs: warp_tiltseries

warp_tiltseries/PositionNNN. refined alignment written back in place
xml

warp_tiltseries/models/ the trained model checkpoints

Watch out

- REFINES an existing coarse alignment — it CANNOT align raw stacks. Run AreTomo -> ts_import_alignments -> select FIRST, or you get a featureless tomogram.
- MA_TRAINING_DEVICES must be a SINGLE GPU. Speed comes from MA_RECON_DEVICES.

Refine alignment (infer)

bash

Reuse a model trained by 'miss-alignment (train)' to align THIS (usually larger) dataset with NO retraining — runs 'miss-alignment infer', loading iterN/model.ckpt for each iteration.

READS

dirs: warp_tiltseries, warp_tiltseries/tiltstack

<MA_MODEL_RUN_DIR>/iterN/model.ckpt models from a finished training run

warp_tiltseries/PositionNNN. reads AND rewrites in place
xml

WRITES

dirs: warp_tiltseries

warp_tiltseries/PositionNNN. refined alignment, no training
xml

Flags MA_MODEL_RUN_DIR MA_INFER_DEVICES 0,1,2,3 MA_START_ITER MA_PREPARE_STACKS 10.0 MA_CONDA_ENV miss-alignment

Watch out

- The larger set still needs its OWN coarse alignment imported first.

Import alignments

WarpTools ts_import_alignments

Imports AreTomo .xf/.tilt alignments back into Warp.

READS

dirs: aretomo_output, warp_tiltseries

aretomo_output[-vN]/Imod/PositionNNN/PositionNNN.xf AreTomo's alignment

warp_tiltseries/PositionNNN. the series to update
xml

WRITES

dirs: warp_tiltseries

warp_tiltseries/PositionNNN. tilt axis + shifts written INTO the xml
xml

Flags --settings warp_tiltseries.settings --alignments aretomo_output/Imod/ --alignment_angpix 1.57

Watch out

- Points at the NEWEST aretomo_output*/Imod/ by default. If that run was killed it may be INCOMPLETE -> 'Could not find PositionNNN.xf' for most series. Type a specific version (e.g. aretomo_output-v2/Imod/) to override.
- A FAILED import DESELECTS every failed series. Re-enable with sync_selection -> Select.

Sync selection

WarpTools `change_selection`

Deselects tilt series with no AreTomo alignment so ts_reconstruct won't crash trying to reconstruct them.

READS

dirs: warp_tiltseries

warp_tiltseries/PositionNNN. the selection flag lives here
xml

Flags `--settings` warp_tiltseries.settings `--input_data`

WRITES

dirs: warp_tiltseries

warp_tiltseries/PositionNNN. sets selected/deselected in place
xml

Watch out

- ANY failed per-item run DESELECTS that item, and downstream steps then SILENTLY SKIP it. After any partial failure, run this with `--select`.
- WarpTools needs EXACTLY ONE mode flag (`--select` / `--deselect` / ...).

6. CTF

Defocus handedness

WarpTools `ts_defocus_hand`

Checks (and optionally flips) defocus handedness. Exactly one mode runs per invocation.

READS

dirs: warp_tiltseries

warp_tiltseries/PositionNNN. CTF fits
xml

Flags `--settings` warp_tiltseries.settings

WRITES

dirs: warp_tiltseries

warp_tiltseries/PositionNNN. sets the defocus handedness
xml

Watch out

- Run `--check` first; only then `--set_*`. Exactly ONE mode flag.

CTF estimation

WarpTools `ts_ctf`

Per-tilt CTF refinement across each series.

READS

dirs: warp_tiltseries, warp_tiltseries/tiltstack

warp_tiltseries/PositionNNN. series metadata
xml

warp_tiltseries/tiltstack/*/ the stacks to fit
*.st

Flags `--settings` warp_tiltseries.settings `--range_high` 7 `--defocus_max` 8 `--device_list` 0 `--perdevice` 2

WRITES

dirs: warp_tiltseries

warp_tiltseries/PositionNNN. CTF written INTO the xml: <CTF><Param Name="Defocus"> (per-series mean, um), DefocusDelta/DefocusAngle (astigmatism), and <GridCTF> (per-TILT defocus, one Node per tilt)

Watch out

- No new files — it EDITS the per-series xml in place.

7. RECONSTRUCT

Tomogram reconstruction

WarpTools `ts_reconstruct`

Back-projects aligned, CTF-corrected tilts into 3D tomograms.

READS

dirs: warp_tiltseries

warp_tiltseries/PositionNNN.xml alignment + CTF

warp_tiltseries/tiltstack/*/*.st the stacks

Flags `--settings` warp_tiltseries.settings `--angpix` 10 `--device_list` 0 `--perdevice` 1 `--deconv` `--dont_invert` on

Watch out

- is the pixel size to TWO DECIMALS: `--angpix 10` -> '10.00Apx'.
- Leaving `--angpix` blank means NATIVE (1.57 Å) -> ~260x the volume, tens of GB and ~40 min EACH. Use ~10 Å for viewing/picking.
- Needs `WARP_FORCE_MRC_FLOAT32=1` for IMOD/3dmod to read the output.

WRITES

dirs: warp_tiltseries/reconstruction

warp_tiltseries/reconstruction/PositionNNN_<apx>Apx.mrc the full tomogram, e.g. Position042_12.56Apx.mrc

8. PICK

Template matching

WarpTools `ts_template_match`

CTF-aware template matching to locate particles (apoferritin example values — adapt per target).

READS

dirs: warp_tiltseries/reconstruction

warp_tiltseries/reconstruction/PositionNNN_<apx>Apx.mrc the FULL TOMOGRAM at `--tomo_angpix`. It REUSES this; it does not rebuild it.

warp_tiltseries/template/emd_<code>.mrc the template (auto-downloaded by `--template_emdb` or your `--template_path`)

warp_tiltseries/PositionNNN.xml alignment + CTF

WRITES

dirs: warp_tiltseries/matching

warp_tiltseries/matching/PositionNNN_<apx>Apx<SUFFIX>.star THE PEAK LIST. <SUFFIX> = `--override_suffix` verbatim if set, else the template-derived name (`_emd_70905 / _<template stem>`)

warp_tiltseries/matching/PositionNNN_<apx>Apx<TEMPLATE>_corr.mrc correlation volume — named after the TEMPLATE, NOT your `override_suffix`

warp_tiltseries/matching/PositionNNN_<apx>Apx<TEMPLATE>_angleid.mrc best-orientation map (same template-based naming)

warp_tiltseries/template/emd_<code>.mrc the downloaded EMDB map

Flags `--settings` warp_tiltseries.settings `--tomo_angpix` 10 `--template_emdb` `--template_path` `--override_suffix` `--subdivisions` 3 `--template_diameter` `--symmetry` C1 `--whiten` on `--optimize_poses` `--check_hand` 2 `--npeaks` 2000 `--peak_distance` `--max_missing_tilts` 2 `--subvolume_size` 192 `--device_list` `--perdevice` 1

Watch out

- `--tomo_angpix` MUST equal an `--angpix` you ALREADY ran `ts_reconstruct` at, or every series fails with 'A reconstruction at the desired resolution was not found.'
- `--override_suffix` renames ONLY the .star. The `_corr.mrc` / `_angleid.mrc` keep the TEMPLATE name — so a viewer needs `-mp '*.star'` but `-cvp '*'`